SHIMADZU

638-94840
Jan. 2021

LabSolutions™ DB/CS

Instrucition Manual

Installation & Maintenance Guide [LabSolutions TOC Volume]

Read this manual thoroughly before you use the product.
Keep this manual for future reference.

This page is intentionally left blank.

Introduction

Read this Instruction Manual thoroughly before using the product.

Thank you for purchasing this product.

This manual describes the operation and options for this product. Read this manual thoroughly before using the product and operate the product in accordance with the instructions in this manual.

Keep this manual for future reference.

Important

- If the user or usage location changes, ensure that this manual is always kept together with the product.
- If this manual or a product warning label is lost or damaged, immediately contact your Shimadzu representative to request a replacement.
- To ensure safe operation, contact your Shimadzu representative if product installation, adjustment, re-installation (after the product is moved) is required.

Notice

- Information in this manual is subject to change without notice and does not represent a commitment on the part of the vendor.
- Any errors or omissions which may have occurred in this manual despite the utmost care taken in its production will be corrected as soon as possible, although not necessarily immediately after detection.
- All rights are reserved, including those to reproduce this manual or parts thereof in any form without permission in writing from Shimadzu Corporation.
- LabSolutions is a trademark of Shimadzu Corporation.
  Windows is a registered trademark of Microsoft Corporation in the United States and/or other countries.
  Third-party trademarks and trade names may be used in this publication to refer to either the entities or their products/services, whether or not they are used with trademark symbol "TM" or "®".

©2021 Shimadzu Corporation. All rights reserved.

Original version is approved in English.

Indications Used in This Manual

The following symbols are used in this manual:

Indication	Meaning
NOTE	Emphasizes additional information that is provided to ensure the proper use of this product.
Reference	Indicates the location of related reference information.

Mouse and Window Operations

Description	Meaning
Click	Means pointing to an object and pressing the left mouse button once.
Right-click	Means pointing to an object and pressing the right mouse button once.
Double-click	Means pointing to an object and pressing the left mouse button twice.
Text in square brackets [ ]	The names of buttons in a window, and window names, are enclosed in square brackets. Example: Click [OK].
Text in quotation marks " "	Entered numerical values, text, and keyboard key names are indicated enclosed in quotation marks. Example: Press the "N" key while holding down the "Ctrl" key.

Use of Instruction Manuals

Shimadzu TOC analyzers include the LabSolutions TOC instrument control software and the LabSolutions Manager data management software as the LabSolutions system.

Refer to the corresponding instruction manual for details on the operation of LabSolutions DB/CS.

Hint For details on how to operate the TOC analyzer, refer to the instruction manual for your particular model.

Instruction Manual	Description
LabSolutions Manager DB Installation & Maintenance Guide or LabSolutions Manager CS Installation & Maintenance Guide	This manual explains software installation, troubleshooting, and environment settings for the LabSolutions system.
LabSolutions DB/CS Installation & Maintenance Guide [LabSolutions TOC Volume] (This manual)	This manual explains LabSolutions TOC model-specific installation, troubleshooting, and environment settings.
LabSolutions Manager DB System Users Guide or LabSolutions Manager CS System Users Guide	This manual explains how to use the LabSolutions system management function.
LabSolutions DB/CS System Users Guide [LabSolutions TOC Volume]	This manual explains how to use LabSolutions TOC model-specific management functions.

For details on each LabSolutions command and execution method, refer to the provided Help file.

Warranty

Shimadzu provides the following warranty for this product.

1. **Period:**

   Please contact your Shimadzu representative for information about the period of this warranty.

2. **Description:**

   If a product/part failure occurs for reasons attributable to Shimadzu during the warranty period, Shimadzu will repair or replace the product/part free of charge. However, in the case of products which are usually available on the market only for a short time, such as personal computers and their peripherals/parts, Shimadzu may not be able to provide identical replacement products.

3. **Limitation of Liability:**

   (1) In no event will Shimadzu be liable for any lost revenue, profit or data, or for special, indirect, consequential, incidental or punitive damages, however caused regardless of the theory of liability, arising out of or related to the use of or inability to use the product, even if Shimadzu has been advised of the possibility of such damage.

   (2) In no event will Shimadzu's liability to you, whether in contract, tort (including negligence), or otherwise, exceed the amount you paid for the product.

4. **Exceptions:**

   Failures caused by the following are excluded from the warranty, even if they occur during the warranty period.

   (1) Improper product handling

   (2) Repairs or modifications performed by parties other than Shimadzu or Shimadzu designated companies

   (3) Product use in combination with hardware or software other than that designated by Shimadzu

   (4) Computer viruses leading to device failures and damage to data and software, including the product's basic software

   (5) Power failures, including power outages and sudden voltage drops, leading to device failures and damage to data and software, including the product's basic software

   (6) Turning OFF the product without following the proper shutdown procedure leading to device failures and damage to data and software, including the product's basic software

   (7) Reasons unrelated to the product itself

   (8) Product use in harsh environments, such as those subject to high temperatures or humidity levels, corrosive gases, or strong vibrations

   (9) Fires, earthquakes, or any other act of nature, contamination by radioactive or hazardous substances, or any other force majeure event, including wars, riots, and crimes

   (10) Product movement or transportation after installation

   (11) Consumables and equivalent items

   Recording media such as CD-ROMs are considered consumable items.

* If there is a document such as a warranty provided with the product, or there is a separate contract agreed upon that includes warranty conditions, the provisions of those documents shall apply.

Software License Agreement

PLEASE READ THIS AGREEMENT CAREFULLY BEFORE USING THE SOFTWARE.
SHIMADZU Corporation ("SMZ") is willing to license the SMZ software provided herein, together with accompanying documentation (collectively "SOFTWARE") to you only upon the condition that you accept all of the terms and condition contained in this License Agreement. By using the SOFTWARE, you agree to be bound by the terms of this Agreement. If you do not agree to all these terms of this Agreement, promptly return the unused SOFTWARE to the party (either SMZ or its reseller) from whom you acquired it to receive a refund of the amount you paid.

1. License

SMZ grants you a non-exclusive and nontransferable license to use the SOFTWARE subject to the following terms and conditions.

2. Limitation of Use

Except as specifically authorized by SMZ, you may NOT:

(1) Use the SOFTWARE, or permit the SOFTWARE to be used, on more than one computer at any one time.

(2) Copy the SOFTWARE except one (1) archival copy of the SOFTWARE.

(3) Modify, reverse engineer, decompile, disassemble, or create derivative works based upon the SOFTWARE.

(4) Transfer, rent, lease or grant any rights in the SOFTWARE in any form to anyone else.

3. Title and Ownership

This license is not for sale and it may not be assigned or sublicensed to anyone else. Title and all associated intellectual property rights to the SOFTWARE shall remain in SMZ and/or its licensor.

4. Upgrades

You are entitled to receive all official software upgrades for the SOFTWARE that SMZ will release as deemed necessary by SMZ. An upgrade means certain supplemental program modules and/or information for bug fixing and/or updates to the defects and/or failures of the SOFTWARE that are acknowledged or confirmed by SMZ. An upgrade excludes hardware, network, consulting services, third party products, operation and general computer system maintenance. All supplemental program module for upgrades and enhancements furnished to you shall be deemed to be part of the SOFTWARE and subject to the terms and conditions set forth in this Agreement.

5. Limited Warranty

SMZ warrants that for a period of one (1) year from the date of purchase, as evidenced by a copy of the receipt, the media on which SOFTWARE is furnished will be free of defects in materials and workmanship under normal use.

Except for the foregoing, SOFTWARE is provided AS IS. Your exclusive remedy and the entire liability of SMZ and its suppliers under this limited warranty will be, at SMZ's option, repair, replacement, or refund of the Software if reported (or, upon request, returned) to the party supplying the SOFTWARE to you. In no event does SMZ warrant that the Software is error free or that you will be able to operate the SOFTWARE without problems or interruptions.

EXCEPT AS SPECIFIED IN THIS WARRANTY, ALL EXPRESS OR IMPLIED CONDITIONS, REPRESENTATIONS, AND WARRANTIES INCLUDING, WITHOUT LIMITATION, ANY IMPLIED WARRANTY OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE, NONINFRINGEMENT, ARISING FROM A

COURSE OF DEALING, USAGE, OR TRADE PRACTICE, ARE HEREBY EXCLUDED TO THE EXTENT ALLOWED BY APPLICABLE LAW.

6. Limitation of Liability

IN NO EVENT WILL SMZ BE LIABLE FOR ANY LOST REVENUE, PROFIT OR DATA, OR FOR SPECIAL, INDIRECT, CONSEQUENTIAL, INCIDENTAL OR PUNITIVE DAMAGES, HOWEVER CAUSED REGARDLESS OF THE THEORY OF LIABILITY, ARISING OUT OF OR RELATED TO THE USE OF OR INABILITY TO USE SOFTWARE, EVEN IF SMZ HAS BEEN ADVISED OF THE POSSIBILITY OF SUCH DAMAGE. IN NO EVENT WILL SMZ'S LIABILITY TO YOU, WHETHER IN CONTRACT, TORT (INCLUDING NEGLIGENCE), OR OTHERWISE, EXCEED THE AMOUNT YOU PAID FOR SOFTWARE.

7. Termination

This License is effective until terminated. You may terminate this License at any time by destroying all copies of SOFTWARE including any documentation. This License will terminate immediately without notice from SMZ if you fail to comply with any provision of this License. Upon termination, you must destroy all copies of SOFTWARE.

8. General

(1) This Agreement is the entire agreement. If any provision of this agreement is held invalid, the remainder of this agreement shall continue in full force and effect.

(2) This Agreement shall be construed and governed in accordance with the laws of Japan, excluding its conflict of law rules.

(3) The exclusive jurisdiction for any disputes arising out of or in connection with this Agreement shall be Kyoto District Court of Japan.

(4) The invalidity or unenforceability of any provision of this Agreement shall not affect the validity or enforceability of any other provision.

After-Sales Service and Availability of Replacement Parts

■ After-Sales Service

If any problem occurs with this product, perform an inspection and take appropriate corrective action as described in the "LabSolutions Manager DB (CS) System Users Guide".

If the problem persists, or the symptoms are not covered in the troubleshooting section, contact your Shimadzu representative.

■ Replacement Parts Availability

Replacement parts for this product will be available for a period of seven (7) years after the product is discontinued. Thereafter, such parts may cease to be available.

Note, however, that the availability of parts not manufactured by Shimadzu shall be determined by the relevant manufacturers.

Contents

Introduction

Indications Used in This Manual ii
Mouse and Window Operations ii
Use of Instruction Manuals iii
Warranty iv
Software License Agreement v
After-Sales Service and Availability of Replacement Parts vi

1 Setup

1.1 Installation of the Software 3
1.1.1 Installation of LabSolutions Manager 3
1.1.2 Installation of LabSolutions TOC 4
1.1.3 License Authentication 6
1.2 Database Creation 7
1.3 Server Connection Settings 7
1.4 Addition of an Acquisition Controller 7
1.5 Uninstallation of the Software 8

2 Hardware Connection

2.1 Installation of the Instrument Driver 9
2.2 Connection of the TOC Instrument and PC 11

3 System Configuration

3.1 TOC System Configuration 12
3.1.1 Registration of Instruments 12
3.1.2 Editing the PC/Instrument Information 14

This page is intentionally left blank.

1 Setup

This chapter describes the procedure for installing the software.

This system is comprised of a database version (stand-alone configuration) that creates a database for saving data on a local PC and a client server version (client/server configuration) that creates a database for saving data on a server for access by clients over a network.

■ Database Version Setup

When using the database version for LabSolutions TOC, perform setup by following the steps below.

Step 1. Installation of LabSolutions Manager (See 1.1.1)

Step 2. Installation of LabSolutions TOC (See 1.1.2)

Step 3. Database Creation (See 1.2)

■ Client Server Version Setup

Each computer in the client/server configuration should include the following. Perform setup in the order of LabSolutions server, acquisition controller, and then client.

Reference For details on system configuration, refer to the LabSolutions Manager CS Installation & Maintenance Guide.

• LabSolutions Server:	This performs consolidation management of information such as system management information and analysis information.
• Acquisition Controller:	This can perform device control and process analysis by connecting to analytical instruments.
• Client:	This performs, data analysis, and system management. TOC instrument control cannot be performed.

LabSolutions Server

Set up the LabSolutions server and the LabSolutions server terminal service by following the steps below.

Reference For details on the operations in each step, refer to the LabSolutions Manager CS Installation & Maintenance Guide.

Step 1. Installation of LabSolutions Server

Step 2. Installation of LabSolutions TOC (See 1.1.2)

Step 3. Database Creation

Step 4. Server Connection Settings

Step 5. Addition of an Acquisition Controller

NOTE
- After "Step 2" described above is applied to LabSolutions server, you can use the TOC instrument.
- Perform "Step 5. Addition of an Acquisition Controller" after completing acquisition controller setup.

Acquisition Controller

Set up the acquisition controller for TOC models by following the steps below.

Step 1. Installation of LabSolutions Manager (Acquisition Controller) (See 1.1.1)

Step 2. Installation of LabSolutions TOC (See 1.1.2)

Client

Set up the client by following the steps below.
If adding a TOC model to a client already set up for another model, perform only Step 2 and Step 3.

Step 1. Installation of LabSolutions Manager (Client) (See 1.1.1)

Step 2. Installation of LabSolutions TOC (See 1.1.2)

Step 3. Server Connection Settings (See 1.3)

1.1 Installation of the Software

This section describes the installation of the software.

This system is comprised of the LabSolutions Manager and LabSolutions TOC, and both of these software must be installed in accordance with the system configuration.

Verify the following items before software installation.

NOTE

- The installer copies the system files to the following folders.
  The software cannot work if the files on these folders are removed, moved or renamed.
  C:\LabSolutions\System
  ("C:\LabSolutions" means target folder for installation.)
  C:\LabSolutions\TOC
  ("C:\LabSolutions" means target folder for installation.)
  C:\Program Files\LabSolutions\TOC
  ("C:" means Windows system drive.)

1.1.1 Installation of LabSolutions Manager

Install the LabSolutions Manager.

The software is comprised of a database edition (stand alone configuration) that creates a database for saving data on a local PC and a client server edition that creates a database for saving data on a server for access by clients over a network.

Install a version of LabSolutions Manager appropriate to your system configuration.

Reference For details, refer to the LabSolutions Manager CS or LabSolutions Manager DB (Installation & Maintenance Guide) instruction manuals.

1.1.2 Installation of LabSolutions TOC

Install the LabSolutions TOC.

1 **Insert the setup disk for LabSolutions TOC into the disk drive.**

The AutoRun window is displayed.

> **NOTE** The disk drive is E in the following example. Use the correct disk drive where "e:\" appears in the example.

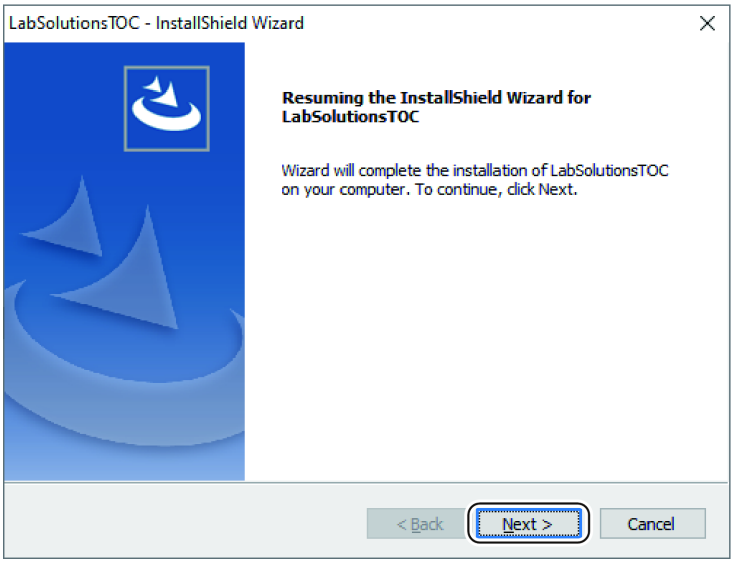
2 **Click [Run Setup.exe], and proceed to Step 5. In other case, proceed to Step 3.**

3 **Click the Start Menu.**

4 **Enter "e:\Setup.exe", and press the "Enter" key.**

The start window is displayed.

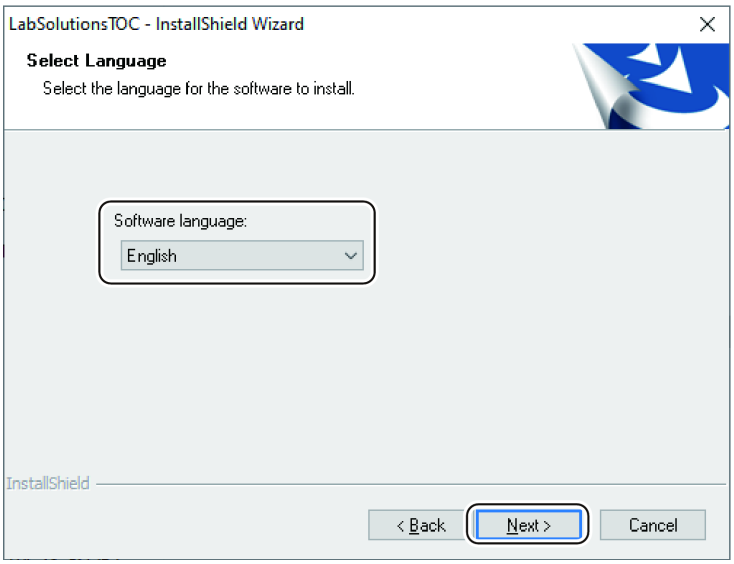
5 **Click [Next].**



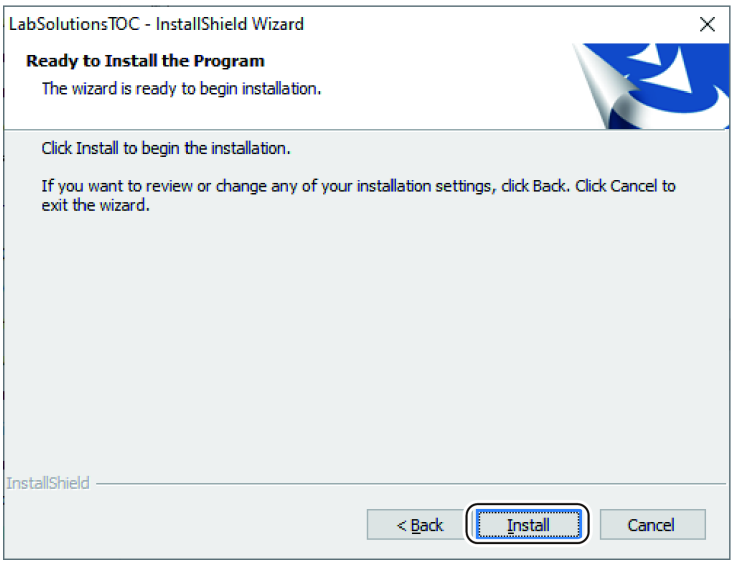
The [Select Language] window is displayed.

6 Select the language to use, then click [Next].

NOTE The language selected here will be displayed in the LabSolutions TOC measurement application.

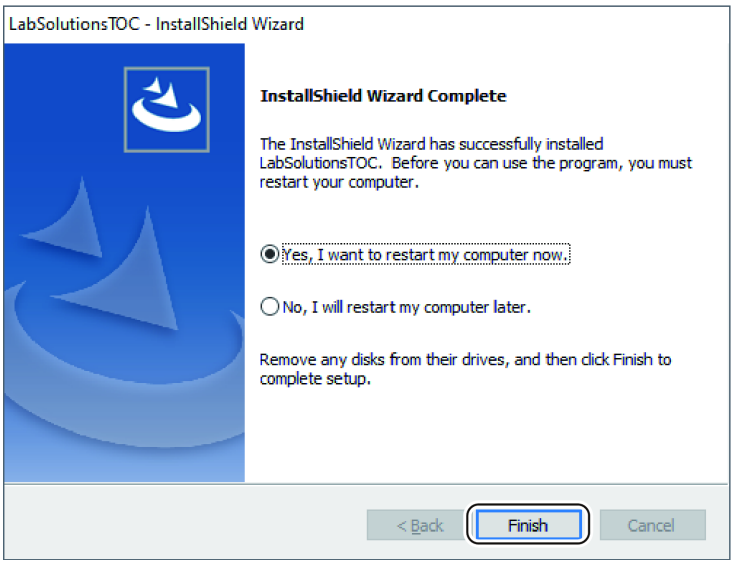


7 Click [Install].



Installation starts.

8 **After installation is completed, select [Yes, I want to restart my computer now.]and then click [Finish].**



This completes the installation process for LabSolutions TOC.

1.1.3 License Authentication

Activate the LabSolutions TOC license.

1 **Click the Start menu.**

2 **In [Shimadzu License Management Tools], select and launch [Software License Management Tools].**

3 **Activate your license according to the instruction manual included with your license certificate.**

The (TOC icon) appears on the [Instrument] tab of the [LabSolutions Main] window.

1.2 Database Creation

For details on database creation, refer to the LabSolutions Manager CS Installation & Maintenance Guide or the LabSolutions Manager DB Installation & Maintenance Guide.

1.3 Server Connection Settings

In a client/server configuration, database connection settings must be configured on clients and the PC installed with LabSolutions server.

For details on these settings, refer to the LabSolutions Manager CS Installation & Maintenance Guide.

1.4 Addition of an Acquisition Controller

For details on adding and changing acquisition controllers (client server version only), refer to the LabSolutions Manager CS Installation & Maintenance Guide.

1.5 Uninstallation of the Software

This section describes the procedure to uninstall the software.

NOTE Back up data files and method files before uninstalling the software.

1 Click [Programs and Features] in the Control Panel.

2 Uninstall [LabSolutions TOC].

1. Select [LabSolutions TOC], and click [Uninstall].
   Uninstallation of LabSolutions UV-Vis starts.
   After uninstallation is finished, the [Uninstall Complete] window is displayed.
2. Click [Finish].

3 Uninstall [SkyPDF Server MT 20xx].

Select [SkyPDF Server MT 20xx], and click [Uninstall].

4 Uninstall [LabSolutions Manager].

1. Select [LabSolutions Manager], and click [Uninstall].
   Uninstallation of LabSolutions Manager starts.
   After uninstallation is finished, the [Uninstall Complete] window is displayed.
2. Click [Finish].
   This completes the uninstallation process.

2 Hardware Connection

This chapter describes the procedure for connecting the TOC analyzer to the PC on which the software is installed.
First connect a USB cable to the PC and the TOC analyzer, and then turn ON the instrument.

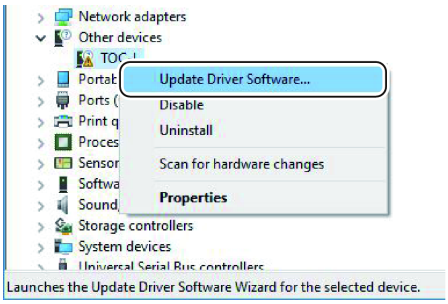
2.1 Installation of the Instrument Driver

1 **Click the Start menu.**

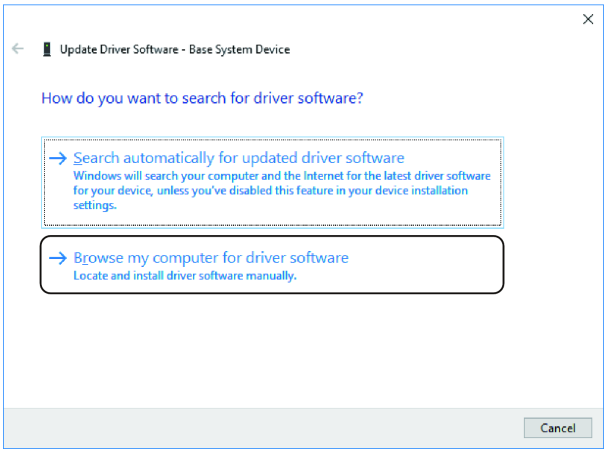
2 **Enter "Device Manager", and press the "Enter" key.**

Hint When the PC recognizes the TOC analyzer, [TOC-XX] is displayed under [Other devices].

3 **Right-click on [TOC-XX], then click [Update Driver Software].**

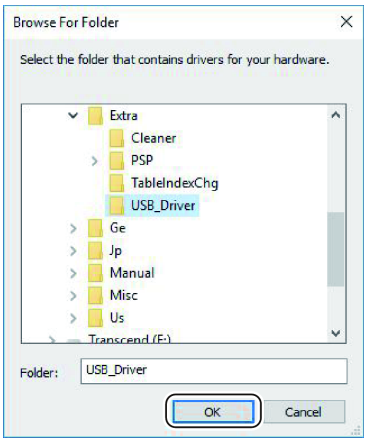


4 **Click [Browse my computer for driver software].**

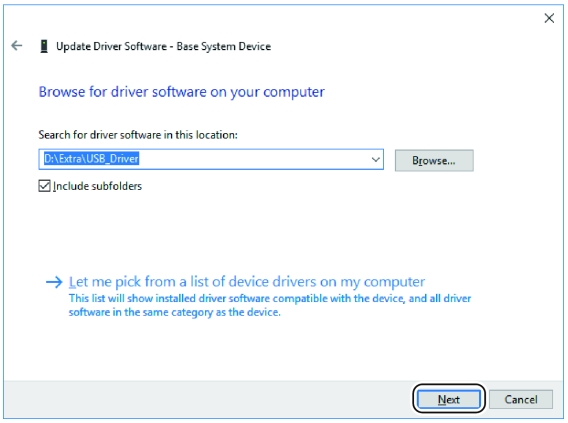


5 **Click [Browse].**

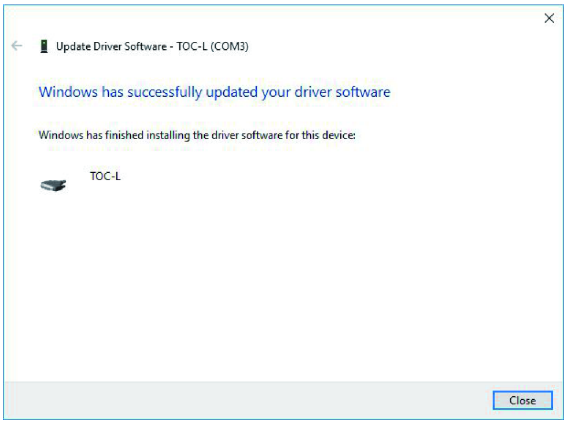
6 **Specify the [Extra] - [USB_Driver] folder on the disk where the software is installed, and then click [OK].**



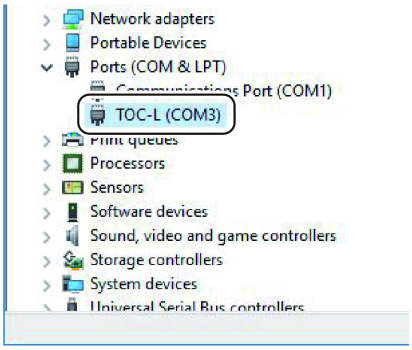
7 **Click [Next].**



Driver software installation starts.
The window shown below is displayed when the installation is completed properly.



NOTE "Check that [TOC-XX] has been added as an item under [Ports (COM & LPT)] in Device Manager.
The displayed COM port number for this item is the port number used for the connection between the instrument and the software."



2.2 Connection of the TOC Instrument and PC

With the TOC model, one TOC instrument can be connected to one PC.
For details, refer to the instruction manual for the instrument.

3 System Configuration

3.1 TOC System Configuration

3.1.1 Registration of Instruments

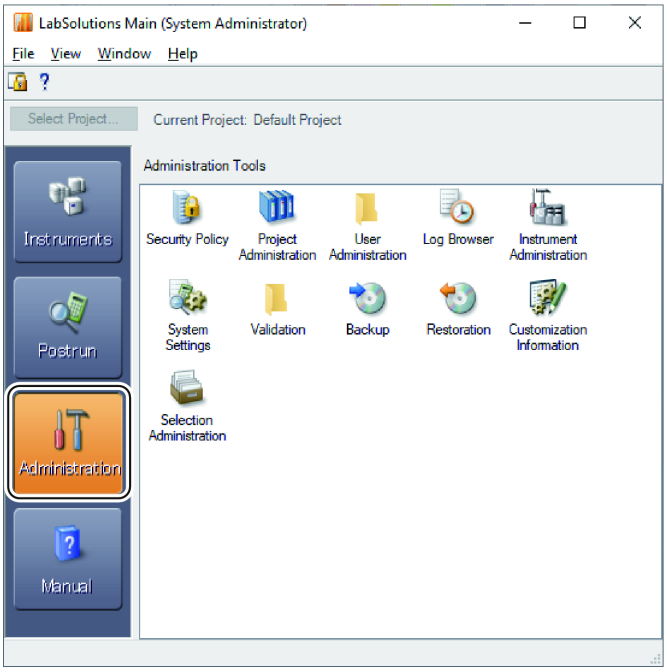
In a client/server configuration, the TOC instrument for use with the LabSolutions system must be registered after installation of this software and addition of an acquisition controller. In the database version, installing this software performs registration automatically.

With the TOC model, one TOC instrument can be connected to one PC.

1 **Double-click on the (LabSolutions) icon and enter a user ID and password.**

The [LabSolutions Main] window is displayed.
If the [Select Project] window is displayed, select a project from the list and click [OK].

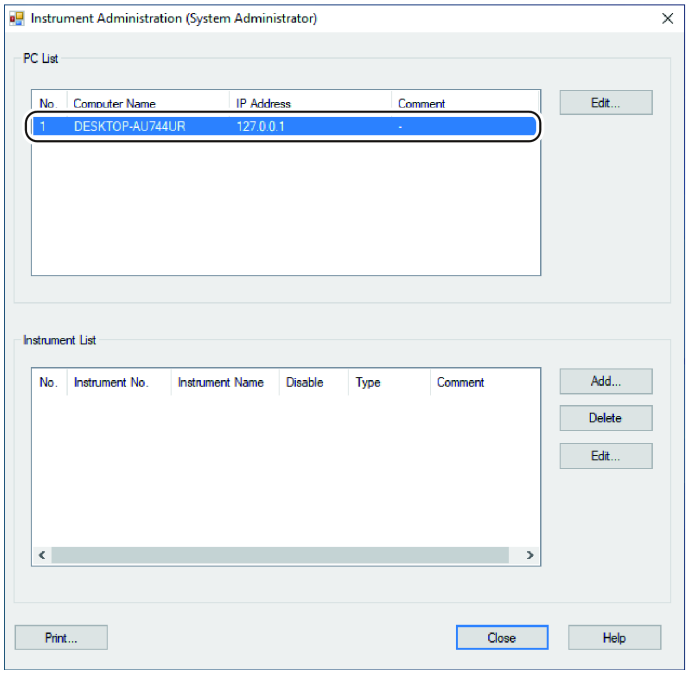
2 **Click [Administration].**



3 **Double-click (Instrument Administration).**

The [Instrument Administration] window is displayed.
Instrument registration is not required when using an instrument that is already registered.
Only register instruments when an instrument is not configured.

4 **If using the client/server version, select the instrument and connected PC from the [Acquisition Controller List].**



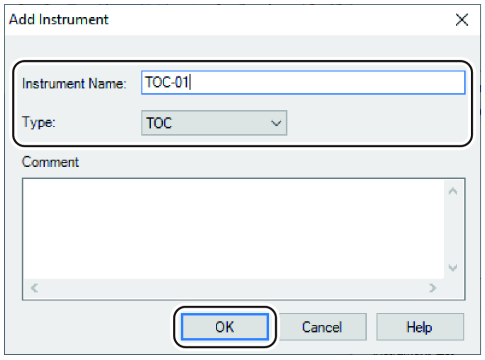
5 **Click [Add] located next to [Instrument List].**

The [Add Instrument] window is displayed.

6 **Enter an appropriate name for [Instrument Name], select [TOC] from the [Type] list, and click [OK].**

The instrument is added to [Instrument List]. Only one instrument can be registered.

Hint In Instrument Name, enter the name used on the system or control number used to distinguish it from the other instruments. If it is not named, enter the model name of the instrument.



The [Instrument Administration] window is displayed again.

7 **Click [Close].**

8 **Click [Instruments] in the [LabSolutions Main] window.**

Selecting [TOC] as the [Instrument Type] displays the instrument that was just added.

3.1.2 Editing the PC/Instrument Information

The PC information is automatically set based on the system information that is acquired during installation. In the database version, the instrument information is also automatically set during installation.
The LabSolutions system allows this information to be edited after installation.

Reference For details, refer to "3.1 Setting the System Information" in the LabSolutions Manager CS System Users Guide or LabSolutions Manager DB System Users Guide.